

# David Le

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## EDUCATION

University of Massachusetts Boston

Expected May 2027

B.S. Computer Science

Relevant coursework: Applied Machine Learning, Software Engineering, Database Management, Computer Security, Advanced Data Structures & Algorithms

## SKILLS

AI / LLM: RAG Pipeline Design, Prompt Engineering (chain-of-thought, few-shot, role prompting), LLM Evaluation, LLM-as-Judge, Hybrid Retrieval (BM25 + dense), Gemini, Ollama

ML / Data: XGBoost, scikit-learn, PyTorch, SentenceTransformers, ChromaDB, FAISS, Pandas, Feature Engineering, A/B Testing, Statistical Evaluation

Backend / APIs: Python, FastAPI, Flask, Node.js, SQLAlchemy, SQL, Gmail API, Firebase

Frontend: React, React Native, Expo, Three.js, JavaScript, HTML, CSS

Other: Git, Java, C, Metasploit, Aircrack-ng, Hugging Face Spaces, Vercel, Docker

## PROJECTS

Ez-Ride

Streamlit, Gemini, BM25, FAISS

Transportation RAG Assistant — UMass Boston

- Built a RAG assistant for campus transit questions using hybrid BM25 + FAISS retrieval, Gemini generation, and source-grounded evaluation workflows to improve answer relevance.

AI Sustainable Labs

Flask, Gemini, ChromaDB, BM25, SentenceTransformers

Public- and Donor-Facing RAG Assistant — UMass Boston SSL

- Built a retrieval-augmented chatbot for the UMass Boston Sustainable Solutions Lab with hybrid retrieval, streaming responses, and evaluation dashboards to monitor answer quality and regressions.

ClaimSight

Python, FastAPI, Gemini, SAM 2

Insurance Damage Assessment Tool

- Built a vehicle damage assessment tool that analyzes uploaded photos and generates structured claim outputs using FastAPI, Gemini summaries, and SAM 2 segmentation with image safety checks and rate limiting.

F1 Race Strategy Predictor

XGBoost, scikit-learn, FastAPI, React

Machine Learning Prediction Platform

- Predicted F1 race outcomes using blended XGBoost and scikit-learn models trained on historical race, qualifying, tyre, and weather data; deployed via FastAPI + React with automated retraining scripts.

Interactive Avatar Chatbot

JavaScript, Firebase, Gemini

Avatar-Guided Mental Health Support App

- Collaborated with a UMass Boston professor to build a multilingual mindfulness app for older Korean adults; implemented Firebase auth, Gemini-powered chat flows, and session-aware avatar interactions.

Email Expense Tracker

FastAPI, Gmail API, Ollama, SQLAlchemy

AI-Powered Expense Automation Tool

- Automated receipt parsing from Gmail using LLM extraction and SQLAlchemy-backed storage; built Firebase session handling and resilient API fallback logic for reliable personal finance tracking.

AI Prompt Engineering Research

Python, Gemini, Hugging Face

LLM Output Quality Research

- Designed an abstract-generation evaluation workflow using chain-of-thought, few-shot, and structured output prompting in Gemini; scored outputs via preference-rate and rubric-based metrics on Hugging Face.

## EXPERIENCE

Server — Gyu-Kaku Japanese BBQ

July 2024 – Present · Boston, MA

Youth Leader — VietAID

July – Sept 2022 · Boston, MA

Serve 40+ guests per shift averaging \$1,200 in sales; strong communication and multitasking under pressure.

- Designed and administered mental health surveys for 200+ Boston Public School students; analyzed data and presented findings to school officials.

**IRS-Certified Tax Preparer — VITA Program**

Sept 2021 – May 2023 · Boston, MA

- Assisted 30+ low-income families with accurate federal tax return preparation.